

QUESTIONS, LABELED BY WHETHER REASONING WAS NEEDED

Reasoning necessary (right with reasoning, wrong without)

Reasoning not needed (right both ways)

OPENTHINKER3-7B, VS ITS NON-REASONING PARENT, LAYER 19

**Predictability of the answer state,  
by whether the question needed reasoning**

Held-out  $R^2$ , context  $\rightarrow$  answer  $\leftarrow$

